Topic 14: Matter – Elements, atoms and compounds

Students should:		Guidance *iLowerSecondary Curriculum reference*
14.1	understand the meaning of the terms *element*, *atom*, *compound* and *molecule*	*Year 7: Matter – Elements, atoms and compounds*
14.2	know Dalton's atomic model	all matter is made of atoms atoms cannot be broken down into anything simpler atoms of the same element are identical and different to atoms of other elements compounds are formed when two or more different atoms join together a chemical reaction involves a rearrangement of atoms *Year 8: Matter – Elements, atoms and compounds*

Students should:		Guidance ***iLowerSecondary Curriculum reference***
14.3	know common chemical symbols and common chemical formulae	limited to the elements: • hydrogen, H • oxygen, O • nitrogen, N • carbon, C • magnesium, Mg • copper, Cu • zinc, Zn • aluminium, Al • iron, Fe • chlorine, Cl • helium, He. the molecules: • hydrogen, H_2 • oxygen, O_2. the compounds: • carbon dioxide, CO_2 • calcium carbonate, $CaCO_3$ • copper carbonate, $CuCO_3$ • calcium chloride, $CaCl_2$ • copper chloride, $CuCl_2$ • magnesium chloride, $MgCl_2$ • sodium chloride, NaCl • zinc chloride, $ZnCl_2$ • copper oxide, CuO • magnesium oxide, MgO • zinc oxide, ZnO • water, H_2O • hydrochloric acid, HCl • sodium hydroxide, NaOH. describe reactions in terms of rearrangement of atoms to form new substance(s)/compound(s) and introduce word equations *Year 8: Matter – Elements, atoms and compounds*
14.4	understand that chemical formulae show the ratio of elements in a compound and be able to use these formulae	*Year 8: Matter – Elements, atoms and compounds*